

DATA ANALYSIS PROJECT

MetroCar SQL

Data Analysis Project

Business Performance • Customer Behavior • Revenue Analysis • Service Optimization

PREPARED BY

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TOOLS

PostgreSQL | pgAdmin 4 | SQL

Project Overview

ABOUT METROCAR

MetroCar is a ride-hailing company seeking to improve customer acquisition, operational efficiency, revenue generation, and customer satisfaction.

PROJECT SCOPE

This project uses SQL to analyze customer behavior, ride operations, transactions, and customer feedback to generate actionable business insights and recommendations.

DATA SOURCES

5

Core operational tables

ANALYTICAL QUESTIONS

10

Business questions answered

ANALYSIS FOCUS

4

Acquisition · Operations · Revenue · CX

Project Objectives

01

Measure conversion from app downloads to user signups.

02

Identify customer funnel drop-off points.

03

Analyze ride cancellations and incomplete trips.

04

Determine peak demand periods.

05

Evaluate ride completion performance.

06

Assess revenue generation and revenue trends.

07

Analyze customer ratings and reviews.

08

Identify drivers of customer satisfaction.

09

Generate actionable business recommendations.

Database Structure

TABLES USED

▣ app_downloads

▣ signups

▣ ride_requests

▣ transactions

▣ reviews

DATA VERIFICATION



SQL

```
SELECT *  
FROM table_name  
LIMIT 10;
```

Each table was inspected with a sample query to confirm schema, data types, and row availability before deeper analysis.

App Downloads → User Signups

SQL QUERY

SQL

```
SELECT
  (SELECT COUNT(*) FROM signups) AS total_signups,
  (SELECT COUNT(*) FROM app_downloads) AS total_downloads,
  ROUND(
    (SELECT COUNT(*) FROM signups) * 100.0 /
    (SELECT COUNT(*) FROM app_downloads),
    2
  ) AS signup_conversion_rate;
```

RESULTS

Total Downloads	23,608
Total Signups	17,623
Conversion Rate	74.65%
FINDING	
Approximately 74.65% of users who downloaded the app completed registration.	
BUSINESS INSIGHT	
The onboarding process is performing well and demonstrates strong customer interest.	
RECOMMENDATION	
Continue optimizing signup and onboarding experiences to improve conversion further.	

Customer Funnel Drop-off

SQL QUERY

SQL

```
SELECT
  COUNT(*)           AS requested_rides,
  COUNT(accept_ts)   AS accepted_rides,
  COUNT(pickup_ts)    AS picked_up_rides,
  COUNT(dropoff_ts)   AS completed_rides
FROM ride_requests;
```

RESULTS

Requested Rides	48,700
Accepted Rides	31,452
Picked-Up Rides	28,416
Completed Rides	28,416

FINDING

The largest drop-off occurs between ride request and ride acceptance.

BUSINESS INSIGHT

Driver availability and ride matching appear to be major bottlenecks.

RECOMMENDATION

Improve dispatch efficiency and increase driver supply.

Ride Cancellations & Incomplete Trips

SQL QUERY

SQL

```
SELECT
  COUNT(*)           AS total_requests,
  COUNT(cancel_ts)    AS cancelled_rides,
  ROUND(
    COUNT(cancel_ts) * 100.0 / COUNT(*),
    2
  ) AS cancellation_rate
FROM ride_requests;
```

RESULTS

Total Requests	48,700
Cancelled Rides	20,284
Cancellation Rate	41.65%

FINDING

Ride cancellations account for a substantial proportion of ride requests.

BUSINESS INSIGHT

High cancellation levels may negatively impact customer experience and revenue.

RECOMMENDATION

Investigate root causes of cancellations and strengthen driver allocation processes.

Peak Ride Demand Periods

QUERY A · PEAK HOURS

●●●

SQL

```
SELECT
  EXTRACT(HOUR FROM request_ts) AS request_hour,
  COUNT(*) AS total_requests
FROM ride_requests
GROUP BY request_hour
ORDER BY total_requests DESC;
```

RESULTS

9 AM	7,663
8 AM	7,579
5 PM	7,435
Sunday	7,022
Tuesday	7,014

QUERY B · PEAK DAYS

●●●

SQL

```
SELECT TO_CHAR(request_ts, 'Day') AS day_of_week,
  COUNT(*) AS total_requests
FROM ride_requests
GROUP BY day_of_week ORDER BY total_requests DESC;
```

FINDING

Demand peaks during commuting hours and remains consistently high throughout the week.

RECOMMENDATION

Align driver scheduling with peak demand periods.

Ride Completion Rate & Revenue Impact

QUERY A · COMPLETION RATE

●●●

SQL

```
SELECT
  COUNT(*)           AS total_requests,
  COUNT(dropoff_ts)  AS completed_rides,
  ROUND(
    COUNT(dropoff_ts) * 100.0 / COUNT(*), 2
  ) AS completion_rate
FROM ride_requests;
```

QUERY B · REVENUE FROM COMPLETED RIDES

●●●

SQL

```
SELECT COUNT(DISTINCT r.ride_id) AS completed_rides,
       ROUND(SUM(t.purchase_amount_usd)::numeric, 2) AS
revenue_generated
FROM ride_requests r JOIN transactions t ON r.ride_id = t.ride_id
WHERE r.dropoff_ts IS NOT NULL AND t.charge_status = 'Approved';
```

RESULTS

Completion Rate	58.35%
Completed Rides	28,416
Revenue Source	Approved

FINDING

Only completed rides generate revenue.

BUSINESS INSIGHT

Increasing completion rates directly increases revenue potential.

RECOMMENDATION

Reduce cancellations and incomplete trips.

Customer Ratings & Service Quality

QUERY A · AVERAGE RATING



SQL

```
SELECT
  ROUND(AVG(rating), 2) AS average_rating
FROM reviews;
```

RESULTS

Average Rating	3.05 / 5
Range	1 – 5

QUERY B · RATING DISTRIBUTION



SQL

```
SELECT rating, COUNT(*) AS total_reviews
FROM reviews
GROUP BY rating
ORDER BY rating;
```

FINDING
Customer experiences are mixed, with both high and low ratings appearing frequently.

RECOMMENDATION
Improve service consistency.

Patterns Linked to Lower Satisfaction

SQL QUERY

SQL

```
SELECT
  CASE
    WHEN r.cancel_ts IS NOT NULL THEN 'Cancelled'
    WHEN r.dropoff_ts IS NOT NULL THEN 'Completed'
    ELSE 'Incomplete'
  END AS ride_status,
  ROUND(AVG(rv.rating), 2) AS avg_rating
FROM ride_requests r
JOIN reviews rv ON r.ride_id = rv.ride_id
GROUP BY ride_status;
```

RESULTS

Completed Rides	3.04
Overall Avg	3.05

FINDING

Service quality remains moderate even among completed rides.

RECOMMENDATION

Focus on reliability and customer experience improvements.

Transaction Revenue Trends Over Time

SQL QUERY

SQL

```
SELECT
  DATE_TRUNC('month', transaction_ts) AS month,
  ROUND(SUM(purchase_amount_usd)::numeric, 2) AS revenue
FROM transactions
WHERE charge_status = 'Approved'
GROUP BY month
ORDER BY month;
```

RESULTS

Time Grain	Monthly
Filter	Approved
Trend	Upward

FINDING

Monthly revenue shows overall business growth over time.

RECOMMENDATION

Continue customer acquisition and retention efforts.

Downloads by Platform

SQL QUERY

SQL

```
SELECT
  platform,
  COUNT(*) AS total_downloads
FROM app_downloads
GROUP BY platform
ORDER BY total_downloads DESC;
```

RESULTS

iOS	14,290
Android	6,935
Web	2,383

FINDING

iOS generates the highest number of downloads.

RECOMMENDATION

Prioritize investment in the iOS user experience.

QUESTION 10

Signups by Age Group

SQL QUERY

SQL

```
SELECT
  age_range,
  COUNT(*) AS total_signups
FROM signups
GROUP BY age_range
ORDER BY total_signups DESC;
```

RESULTS

35 – 44	5,181
25 – 34	3,447
18 – 24	1,865
45 – 54	1,826

FINDING

Users aged 35–44 represent the largest signup segment.

RECOMMENDATION

Target marketing campaigns toward this demographic.

Customer Engagement — Reviews on Completed Rides

QUERY A · RIDES WITH REVIEWS

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SQL

```
SELECT
  COUNT(DISTINCT rv.ride_id) AS rides_with_reviews
FROM reviews rv;
```

RESULTS

Review Rate	69.90%
Scope	Completed

QUERY B · REVIEW PARTICIPATION RATE

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SQL

```
SELECT ROUND(
  COUNT(DISTINCT rv.ride_id) * 100.0 /
  COUNT(DISTINCT r.ride_id), 2
) AS review_rate
FROM ride_requests r
LEFT JOIN reviews rv ON r.ride_id = rv.ride_id
WHERE r.dropoff_ts IS NOT NULL;
```

FINDING

Nearly 70% of completed rides generated customer feedback.

RECOMMENDATION

Encourage reviews through reminders and incentives.

Final Key Findings

74.65%

Signup conversion rate

Request → Accept

Largest funnel drop-off

41.65%

Cancellation rate

58.35%

Ride completion rate

Commute hrs

Peak demand window

Upward

Revenue trend over time

3.05 / 5

Average customer rating

iOS

Highest-download platform

35 – 44

Largest signup segment

69.90%

Completed rides reviewed

Final Recommendations

01 Improve driver availability.

02 Reduce ride cancellations.

03 Optimize ride matching.

04 Increase ride completion rates.

05 Encourage customer reviews.

06 Improve service quality consistency.

07 Focus on high-performing customer segments.

08 Use data-driven decision making.

CONCLUSION

Turning Insights into Action

The analysis identified opportunities to improve customer acquisition, ride completion, operational efficiency, revenue generation, and customer satisfaction.

By implementing the recommended actions, MetroCar can strengthen customer experience, improve operational performance, and support long-term business growth.

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